

ECB Monitor Methodology Report: ECB Council Stance

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Abstract

This note describes the methodology underlying the ZEW ECB Monitor, a tool for systematically tracking the monetary policy communication of members of the ECB Governing Council. Drawing on the Bank for International Settlements (BIS) speech database, supplemented by real-time scraping for the most recent period, the monitor classifies each speech on a continuous hawk–dove scale using a large language model (GPT-5 mini). The AI-generated classifications are validated against a human-annotated benchmark of 600 speech paragraphs, yielding a strong correlation between model and human assessments. Individual speech scores are aggregated into three complementary views: a rolling six-week Momentum Indicator capturing real-time shifts in council communication, an annual cross-member comparison revealing the distribution of positions across council members, and a career-average summary of individual hawk–dove orientations. Together, these indicators provide a comprehensive framework for monitoring ECB monetary policy signals.

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1. Project Context

The ECB Monitor was established by ZEW – Leibniz Centre for European Economic Research with support from the Monetary Stability Foundation (“Stiftung Geld und Währung”).

2. Data

The ECB Monitor draws on speeches retrieved from the Bank for International Settlements (2024) speech database, which constitutes the most comprehensive publicly available repository of central bank communications. It covers speeches by executive board members and national governors of the ECB Governing Council from 1998 onwards. For the most recent period, there is no readily available speeches dataset so that the most recent speeches are scraped directly from the BIS website. After speech retrieval, we process the data to extract the raw text from files. For a few of the speeches, an Optical Character Recognition approach to extract the text is necessary. The final speech dataset consists of all available speeches in the BIS database from ECB Governing Council members since the founding of the ECB. Only speeches delivered during a council member’s term in office are retained.

3. Classification

Each speech is submitted in full to a GPT model (currently GPT-5 mini) via the OpenAI API. The model is assigned the system role of a PhD-trained monetary policy researcher to set an appropriate context. A structured prompt asks the model to classify the speaker’s monetary policy stance on a continuous hawk–dove scale from -1.0 to $+1.0$. Concretely, the model is instructed as follows:

“[...] Classify the opinion of the speaker towards the role of central banks to limit the inflation rate and to protect price stability. Return a continuous value with one decimal place from -1.0 to $+1.0$. The interpretation for the extreme and the central values are: -1.0 is appropriate if the speaker currently sees no risks at all that the inflation rate is or will be too high and that there is therefore currently room for a more expansionary use of conventional or unconventional monetary policy instruments; 0.0 is appropriate if the speaker expresses that the current use of monetary policy instruments is exactly right to reach the central bank’s inflation objective; 1.0 is appropriate if the speaker currently sees massive risks that the inflation rate is or will be too high in the future and that there is therefore now the need for a more restrictive monetary policy.[...]”

The hawk–dove position score forms the basis for the “ECB council stance” section in the ECB Monitor.

4. Validation

Before applying the AI-generated classifications we tested the model’s predictions against a human-labelled dataset described in the accompanying research paper (Heinemann and Kemper, 2026). Speeches were segmented into five-sentence parts, and a random sample of 600 paragraphs was drawn from this pool. Each paragraph was independently classified by both the model and a human annotator using the same scoring rubric. The comparison yields a strong

correlation between model and human assessments, providing evidence that the AI classification captures substantive content rather than noise. Nonetheless, users should remain aware of potential misclassification risk, particularly where the number of available speeches per council member or period is small.

5. Council Stance: Different Levels of Aggregation and Different Perspectives

The speech-level position scores are aggregated along three dimensions, each displayed on a dedicated subpage of the ECB Monitor.

5.1 ECB-ZEW Momentum Indicator: Timeline View and Real-Time Information (Six-Week Window)

The Momentum Indicator aggregates position scores across all ECB Governing Council members within rolling six-week windows, computing the arithmetic mean of all scored speeches falling within each window. The resulting time series tracks the evolution of the collective council position and is displayed alongside the ECB deposit rate, allowing users to assess the alignment between communication and policy decisions.

Not all national governors deliver speeches in every six-week window. The indicator therefore reflects the average position only of those members who gave at least one speech during the respective interval and may not be representative of the full council.

The decision to use a six-week window is justified by the aim of the Momentum Indicator, which is to provide real-time information on current trends in ECB communication. Ideally, this information will have predictive power with regard to the next monetary policy decision. The indicators described below opt for covering longer periods in order to provide more substantive information on the heterogeneity within the ECB Council.

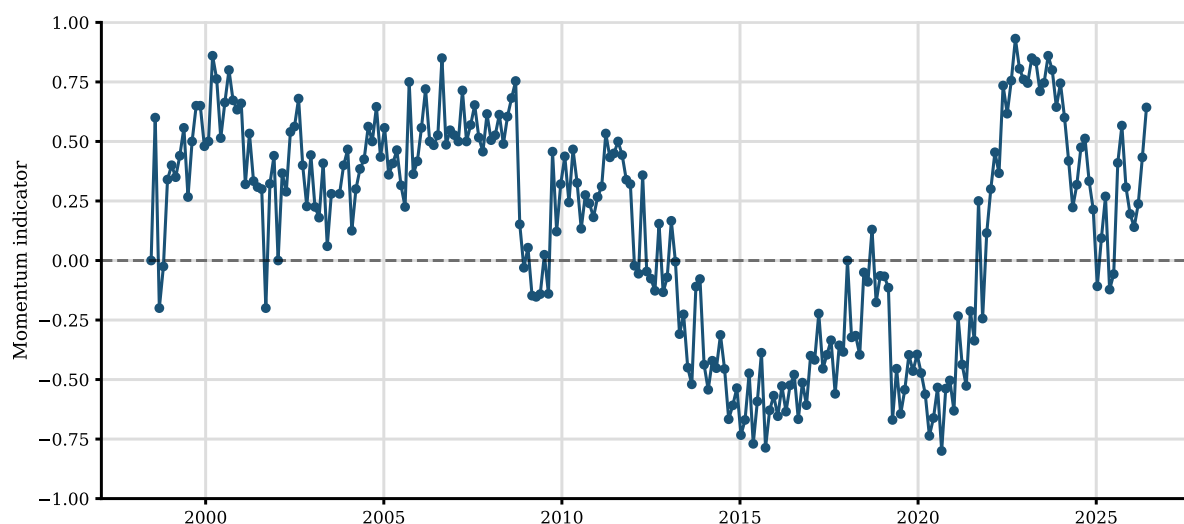


Figure 1: ECB-ZEW Momentum Indicator (six-week rolling windows), 1998–2026. Source: ZEW ECB Monitor, BIS speech database.

Figure 1 illustrates the Momentum Indicator over its full historical range. The blue line shows the six-week rolling hawk–dove score across all scored speeches. The series visualizes how council communication has evolved through the low-inflation decade of the 2010s, the strongly

hawkish shift during the 2022–2023 inflation surge, the subsequent decline as inflation normalized, and the renewed upward movement in 2026. The chart thus serves as an example of how the high-frequency six-week window translates speech-level classifications into a continuous, real-time signal of the collective council stance.

5.2 Majorities in the ECB Council: Individuals and Median (One-Year Window)

For each calendar year, the arithmetic mean position score is computed per council member from all speeches delivered during that year. For the most recent period, the window covers the preceding 12 months on a rolling basis. The scores enable direct cross-member comparisons. An arrow in the visualization indicates the median position across all active members in the selected period. Users may select any year over the past two decades or the trailing 12-month window.

Only council members who delivered at least one scored speech in the respective period are included. The median and cross-member comparisons are therefore conditional on speech activity.

5.3 Career-Aggregate Positions (Full History in Office)

This view presents the arithmetic mean position score computed over each council member's entire speech history since appointment until leaving the ECB Governing Council. It thus provides a long-run summary of individual communication positions and allows comparison of overall hawk–dove orientations across incumbents and former members.

Career-average scores are sensitive to the prevailing inflation environment during a member's tenure. In low-inflation periods, even hawkish-leaning central bankers may have delivered comparatively moderate rhetoric, compressing career scores relative to those accumulated during high-inflation episodes. Scores are therefore best interpreted in conjunction with the timeline view.

References

- Bank for International Settlements (2024). Central bank speeches, 1997–2026. <https://www.bis.org/cbspeeches/download.htm>.
- Heinemann, Friedrich and Kemper, Jan (2026). ECB Council Members' Objectives and Public Debt at Home – Evidence from an AI-Based Textual Analysis. CESifo Working Paper No. 12493. Available at SSRN: <https://ssrn.com/abstract=6301200>.